

**Name:** Neelambika  
**Section:**L  
**SRN:**PES1UG24CS679  
**SUB:** SE

# Use-Case Flow Specification

**Use Case ID:** UC-01

**Use Case Name:** Dispense Medicine with FEFO Enforcement

**Primary Actor:** Pharmacy Clerk

**Secondary Actor:** Hospital Inventory Engine

## Preconditions

1. The Pharmacy Clerk is authenticated and logged into the active terminal session.
2. The requested medicine exists in the system with at least one active, unexpired batch in the database.

## Postconditions

1. Selected medicine batch unit count is decremented in real time.
2. Sales receipt and batch-specific dispensing label are generated.
3. Inventory engine updates live stock and evaluates remaining units against the re-order safety threshold.

## Main Success Scenario (Happy Path)

1. Pharmacy Clerk enters the prescription ID or scans the medicine barcode at the checkout terminal.
2. System queries the inventory database for all active batches associated with the requested medicine.
3. System filters out expired records and sorts unexpired batches in ascending order of expiry date (**FEFO Order**).
4. System automatically selects and reserves units from the batch with the nearest valid expiry date.
5. Pharmacy Clerk inputs the requested dispensing quantity.
6. System verifies the chosen batch contains sufficient quantity to fulfill the request.
7. System updates the live inventory ledger, deducts the stock, and displays the transaction summary.
8. Pharmacy Clerk confirms the checkout and completes payment collection.
9. System generates the sales invoice and prints the batch dispensing slip.
10. Use case ends successfully.

## Alternate Flows

- **3a. Batch Within Near-Expiry Quarantine Window (<=30 Days):**
  - 3a1.** System flags the batch status as `QUARANTINED_NEAR_EXPIRY`.
  - 3a2.** System blocks the batch from customer dispensing and routes it to the return/disposal queue.
  - 3a3.** System automatically evaluates the next available batch in the FEFO list and resumes at Step 4.
- **6a. Insufficient Quantity in Single Batch (Split-Batch Pick):**
  - 6a1.** System allocates all available units from the nearest expiring batch.
  - 6a2.** System allocates the remaining required balance from the next earliest expiring batch.
  - 6a3.** System prompts the Clerk with a multi-batch split-pick alert and resumes at Step 7.